URP Anime Grass Interaction System

Technical Documentation — Shader + Controller Setup Guide

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1. Overview

This package contains a stylized (anime-style) grass shader for Unity's Universal Render Pipeline (URP), along with a third-person character controller that drives the grass interaction in real time. Grass blades bend and push away from the player as they move through the field, and sway naturally with wind. The system works together using two files:

- **URP_Grass.shader** — the grass shader (wind sway + player push/bend, toon-style shading, shadow/depth support).
- **AnimeGrassController.cs** — third-person player controller that feeds the player's position to the shader every frame.

2. Requirements

- Unity 2022 LTS or newer (Unity 6 recommended)
- Universal Render Pipeline (URP) installed and set as the active pipeline
- Unity's new **Input System** package installed (the controller script uses UnityEngine.InputSystem)
- A grass mesh (plane/blade mesh) with a UV-mapped alpha texture

3. Setup Instructions

Step 1 — Import the Shader

Copy **URP_Grass.shader** into your project's Assets folder. Create a new Material, and set its shader to **Nyxis / URP_Grass** from the shader dropdown.

Step 2 — Assign the Material

Assign your grass texture to the **Alpha Texture** slot on the material (RGB = tint, Alpha = blade cutout shape). Apply this material to your grass mesh renderer(s).

Step 3 — Set Up the Player

Add a **CharacterController** component to your player GameObject (the script requires it). Then add the **AnimeGrassController.cs** script to the same GameObject. Assign a **Camera Pivot** transform in the inspector (an empty child object near the player's head/shoulder works well).

Step 4 — Ground & Camera Collision Layers

In the inspector, set the **Ground Layer** field to whatever layer your terrain/ground uses (used for jump/grounded checks). Set **Camera Collision Mask** to the layers the camera should not clip through (e.g. Default, Terrain — excluding the Player layer).

Step 5 — Press Play

That's it. The controller automatically sends the player's position to the shader every frame using **Shader.SetGlobalVector("_GlobalPlayerPos")** and **Shader.SetGlobalFloat("_GlobalPlayerRadius")** — no extra wiring needed. Any grass using this shader anywhere in the scene will react to the player automatically.

4. Shader Properties Reference

These properties appear on the material Inspector when **Nyxis/URP_Grass** is selected.

Property	What It Does
Alpha Texture (_MainTex)	Blade texture. RGB is used as color tint, Alpha channel defines the blade cutout shape.
Top (Sun) Color	Color applied to the tip of the grass blade (lit side).
Root (Shadow) Color	Color applied to the base/root of the grass blade.
Shadow Tint	Extra tint multiplied in in areas that are in shadow.
Blade Height	Reference height used to calculate the 0–1 gradient along the blade for coloring and bending.
Flatten Normals	Blends the blade's normal toward straight-up (0,1,0). Higher values give flatter, more toon-like lighting.
Wind Speed	How fast the wind sway animation plays.
Wind Strength	How far the grass sways side to side from wind.
Interaction Power (_PushStrength)	How far grass is pushed sideways away from the player.
Bend Down Strength	How much the grass is pushed downward (flattened) near the player.
Falloff Softness	Controls how sharply the push effect fades out from the center to the edge of the interaction radius.
Alpha Cutoff	Alpha threshold below which pixels are discarded (controls the crispness of the blade cutout edge).

5. Controller Script Settings Reference

These fields appear in the Inspector on the **AnimeGrassController** component.

Setting	What It Does
Walk Speed / Run Speed	Movement speed while walking, and while holding Left Shift to sprint.
Rotation Speed	How quickly the character turns to face the movement direction.
Jump Height / Gravity	Controls jump arc height and fall speed.
Ground Check Offset / Ground Layer	Used to detect whether the player is standing on the ground.
Player Height	Sets the CharacterController's height and center automatically at Start.
Grass Push Radius	How far around the player the grass reacts (feeds _GlobalPlayerRadius to the shader).

Setting	What It Does
Grass Push Smooth Time	Smooths the interaction position so the push effect doesn't snap instantly. Lower = snappier, higher = smoother/laggier.
Camera Pivot	The transform the camera orbits/looks at — usually near the player's shoulder or head.
Mouse Sensitivity	How fast the camera rotates with mouse/right-stick movement.
Distance From Player / Min-Max Zoom	Default and allowed range for camera zoom distance (mouse scroll wheel).
Pitch Min/Max	Limits how far up/down the camera can look.
Cam Smooth Time	Smooths camera rotation for a less jittery feel.
Shoulder Offset / Height Offset	Positions the camera to the side (over-the-shoulder) instead of directly behind the player.
Camera Collision Mask/Radius/Buffer	Prevents the camera from clipping through walls/terrain by pulling it closer when something is in the way.

6. Feature Summary — What This System Can Do

- Real-time grass bending/pushing around the player, with adjustable radius, strength and falloff
- Wind sway animation, independent of player interaction
- Stylized toon-style shading with separate top/root colors and shadow tinting
- Full URP support: works with the Forward, Shadow Caster, Depth Only and Depth Normals passes, so grass correctly casts/receives shadows and works with post-processing (SSAO, etc.)
- GPU instancing support for good performance with large amounts of grass
- Third-person controller with WASD/gamepad movement, sprint, jump, mouse-look camera, zoom, and camera collision so the view doesn't clip through geometry
- Works automatically on all grass instances in the scene — no manual per-object setup needed once the material is applied

7. Customization Tips

- For a bigger reaction area, increase **Grass Push Radius** on the controller and match it visually by adjusting **Falloff Softness** on the material.
- For snappier interaction (e.g. combat games), lower **Grass Push Smooth Time**. For a softer, floaty feel, raise it.
- To change the overall grass color palette, edit **Top Color**, **Root Color** and **Shadow Tint** — no code changes needed.
- To use this on other movable objects (not just the player), you can call **Shader.SetGlobalVector("_GlobalPlayerPos", pos)** and **Shader.SetGlobalFloat("_GlobalPlayerRadius", radius)** from any script — the shader doesn't care what moves it, only the global values.

8. Troubleshooting

Problem	Likely Cause / Fix
Grass doesn't react to player at all	Make sure the player GameObject has AnimeGrassController.cs running, and that the grass material actually uses the Nyxis/URP_Grass shader.
Console error about CharacterController missing	Add a CharacterController component to the same GameObject as the script (it is a required component).
Camera doesn't move with mouse	Check that Unity's Input System package is installed and set as active (Project Settings > Player > Active Input Handling).
Camera clips through walls	Assign the correct layers to Camera Collision Mask, excluding the Player's own layer.
Grass looks fully flat/no shading	Lower the Flatten Normals value on the material for more natural lighting variation.

Nyxis — 3D Game Artist. Free asset, provided as-is for community use.